EXEC_13-230: Fixed-Scale t_N + Core Resolution (EJ-230)

Réplica del análisis EJ-204 de la reunión 12-jun con G. Vásquez

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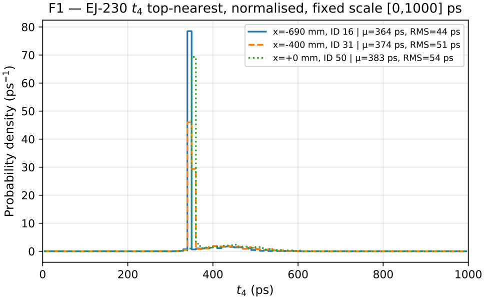
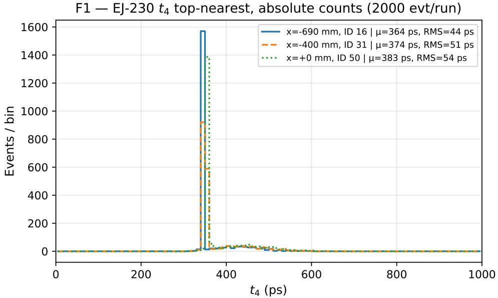
F1 — Overlay t_4, normalised, fixed scale [EJ-230]



How to read this plot

Why: EXEC_12 EJ-204 histograms had auto-scale; this re-bins at 10 ps with fixed [0,1000] ps range for visual comparison between runs. EJ-230 $\tau_d = 1.5$ ns (shorter than EJ-204 1.8 ns). **Axes:** x: t_4 arrival time (ps); y: prob. density (norm, ps^{-1}) or events/10 ps (abs). Note: peak density CAN exceed 1 ps^{-1} — correct. **Takeaway:** Narrow (t4) or broader (t20) distribution; shape encodes scintillation + Landau fluctuations.

F1 — Overlay t_4, absolute counts [EJ-230]



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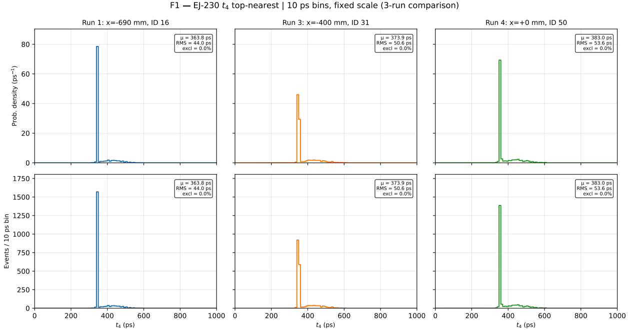
F1 — Individual panels t_4 (identical scale) [EJ-230]



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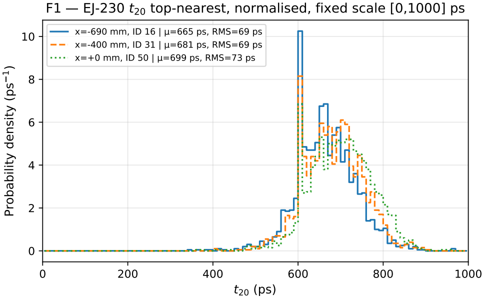
F1 — Overlay t_{20}, normalised, fixed scale [EJ-230]



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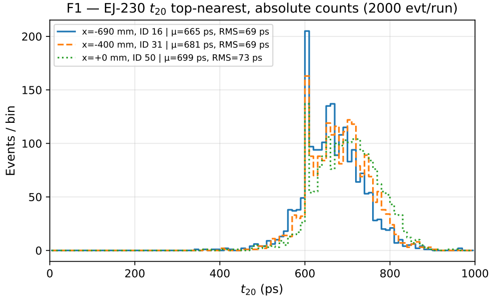
F1 — Overlay t_{20}, absolute counts [EJ-230]



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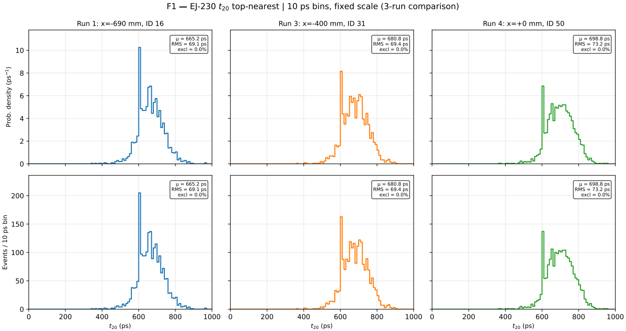

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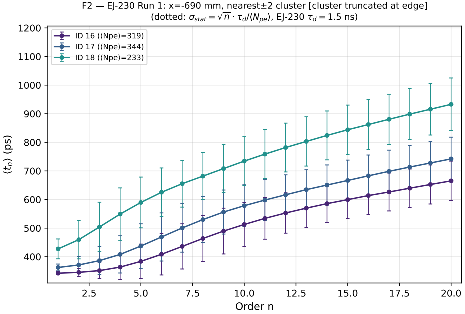
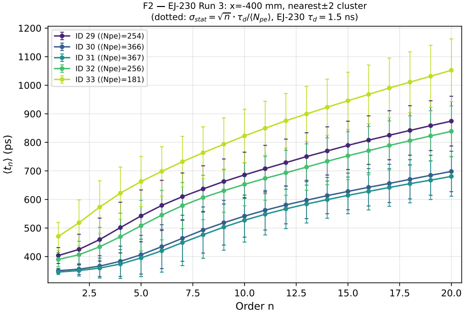
F2 — Run 1: $x = -690\,\mathrm{mm}$ cluster dispersion [EJ-230]



How to read this plot

Why: Do neighbour SiPMs ± 1, ± 2 carry the same timing as the nearest? Stat floor uses EJ-230 $\tau_d = 1.5\,\mathrm{ns}$. **Axes:** x: photon order n (1–20); y: $\langle t_n \rangle$ (ps). Error bars = RMS. Dotted: stat floor $\sqrt{n}\,\tau_d / \langle N_{pe} \rangle$. **Takeaway:** Neighbours approach stat floor when their $\langle N_{pe} \rangle$ is comparable to the nearest.

F2 — Run 3: $x = -400$ mm cluster dispersion [EJ-230]



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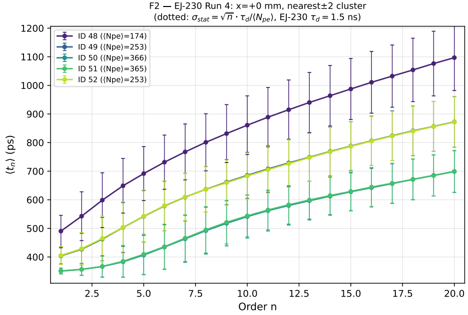
F2 — Run 4: $x = +0\,\mathrm{mm}$ cluster dispersion [EJ-230]



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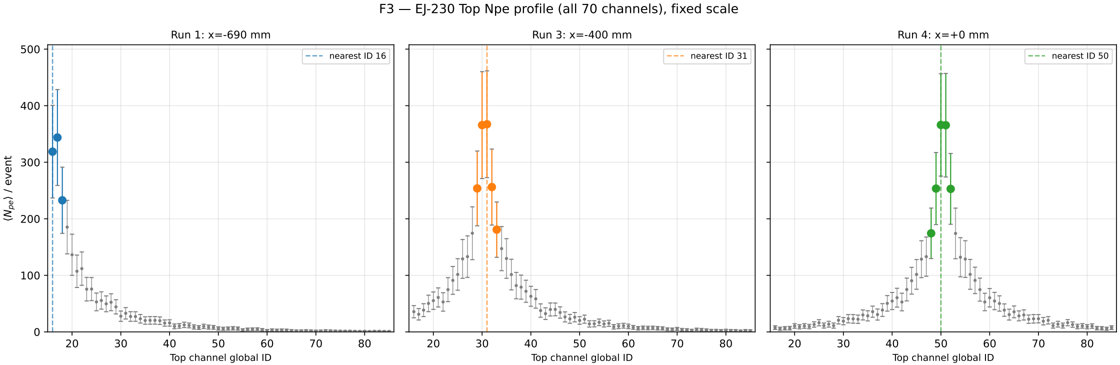
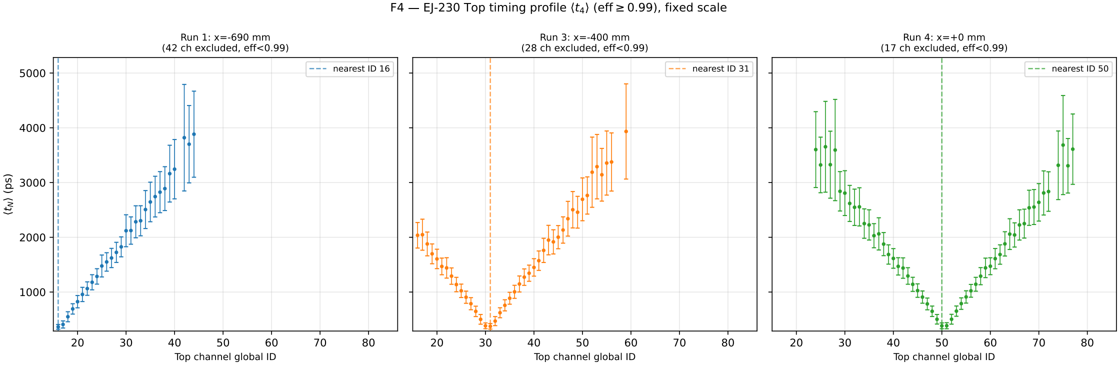
F3 — EJ-230 Top N_{pe} profile (all 70 channels, fixed scale)



How to read this plot

Why: Verify nearest channel is the Npe maximum; shows photon budget across top array. **Axes:** x: global channel ID (16–85); y: $\langle N_{pe} \rangle$/event $\pm$ RMS. Highlighted: cluster ± 2. **Takeaway:** Npe falls off rapidly away from track; ± 2 neighbours still collect useful light.

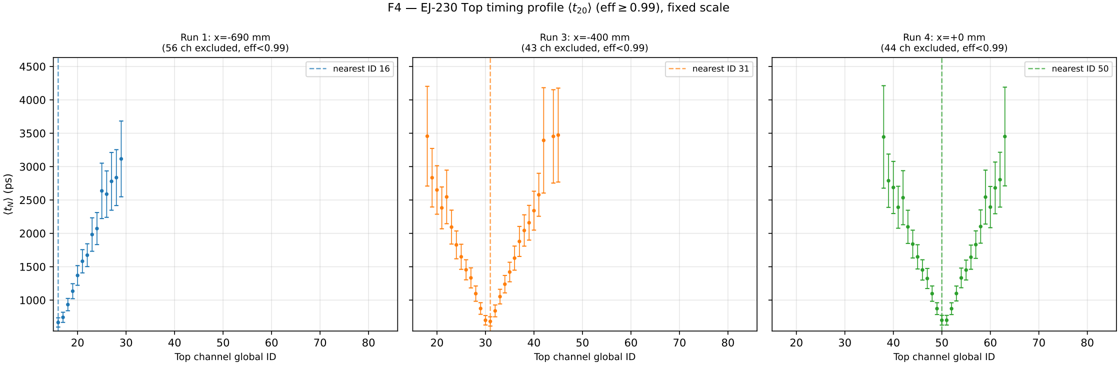
F4 — EJ-230 timing profile $\langle t_4 \rangle$ (eff$\geq$0.99, fixed scale)



How to read this plot

Why: Only channels with eff $\geq$ 0.99 are shown; far channels excluded at N=20 (insufficient Npe) — listed in CSV. **Axes:** x: global channel ID; y: $\langle t_N \rangle$ (ps) $\pm$ RMS. **Takeaway:** Timing rises steeply for channels far from track (longer path + lower N_{pe}).

F4 — EJ-230 timing profile $\langle t_{20} \rangle$ (eff$\geq$0.99, fixed scale)



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Validation gates [EJ-230]

- **G-MAT/G-TAU**: $\tau_d = 1.5\,\mathrm{ns}$ (EJ-230), read from `analysis/exec13/common13.py` — not a literal
- **G1**: nearest IDs determined programmatically from $\langle N_{pe} \rangle$ data: $x = -690 \rightarrow$ ID 16 — $x = -400 \rightarrow$ ID 31 — $x = +0 \rightarrow$ ID 50
- **G2**: all figure panels within each family share identical axis limits — verified post-plot
- **G3**: $n_{\text{events}} = 2000$ per run
- **G4**: $n_{\text{in}} + n_{\text{excl}} + n_{\text{ovfl}} = 2000$ per histogram
- **G5**: F2 x-axis labelled "Order n", range 1–20
- **G-MACRO**: zero `\newcommand` with digits in the command name

All gate results: `exec13_230_gates.csv`. Commit: `79e701df1a3f`.

Provenance and methodology [EJ-230]

Material: EJ-230 (OPSC-106) — $\tau_r = 0.5\,\mathrm{ns}$, $\tau_d = 1.5\,\mathrm{ns}$, yield $= 9700\,\mathrm{ph/MeV}$, $\ell_{\mathrm{att}} = 120\,\mathrm{cm}$, $\lambda_{\mathrm{peak}} = 391\,\mathrm{nm}$.

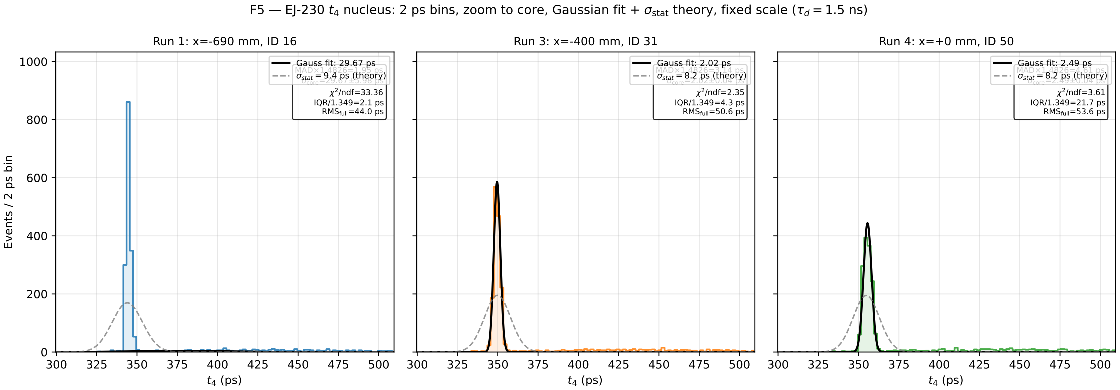
Data: 31-position beam scan, 2000 events/position. Three positions analysed: $x = -690, -400, 0\,\mathrm{mm}$.

Algorithm: `compute_tN` ported unchanged from `exec07/exec12_tN_analysis.py:72--100` (sorted time-to-N-th hit per event). $t = 0$: Geant4 event start (`exec07_photon_budget.py:342`).

No gates applied: no SPTR, no jitter, no SPE convolution, no temporal window.

EXEC_12-EJ230 reconciliation: no published EXEC_12 sigma_fit exists for EJ-230. EXEC_12-style refit (25 ps bins, mean$\pm$2RMS) computed on EJ-230 data for informational reference only.

Analysis script: `analysis/exec13/exec13_230_fixed_scale.py`.



How to read this plot

Why: RMS of t_4 (tens of ps) is dominated by reflection tail, not the prompt core. Zoom to mode$\pm$50 ps with 2 ps bins to measure the narrow nucleus. **Axes:** x: t_4 (ps), zoom $\approx$ mode$\pm$50 ps; y: events/2 ps. Black solid: Gaussian fit. Gray dashed: σ_{stat} theory. **Takeaway:** Narrow Gaussian nucleus ($\sim$ few ps); tail starts at mode+$3\sigma_{\text{core}}$.

Resolución Top nearest: núcleo vs cola (EJ-230, $\tau_d = 1.5\,\mathrm{ns}$)

Posición	MAD×1.4826 (ps)	σ_{core}±err (ps)	χ^2/ndf	IQR/1.349 (ps)	$\mathrm{RMS}_{\mathrm{full}}$ (ps)	σ_{stat} (ps)
$x = -690\,\mathrm{mm}$	1.95	29.67±5.98	33.36	2.13	44.00	9.41
$x = -400\,\mathrm{mm}$	3.14	2.02±0.04	2.35	4.32	50.63	8.17
$x = +0\,\mathrm{mm}$	4.61	2.49±0.04	3.61	21.72	53.61	8.20

Mensaje clave (aprendizaje EJ-204)

Estimador preferido: MAD×1.4826 (robusto, resistente a colas). σ_{core} (ajuste gaussiano 2 ps) tiende a *sub-estimar* con χ^2/ndf elevado — ver tabla. Si IQR anómalamente grande (p.ej. posición central), usar MAD. Cola = $t_4 > \mathrm{modo} + 3\sigma_{\mathrm{core}}$; dominada por reflexiones. σ_{core} y MAD son *intrínsecos de simulación* (sin SPTR, sin jitter); la resolución real estará dominada por el SPTR (≈ 100 ps, bloque EXEC_02b).

Observaciones y próximos pasos [EJ-230]

- **Material EJ-230 vs EJ-204:** $\tau_d = 1.5\,\mathrm{ns}$ (EJ-230) vs $1.8\,\mathrm{ns}$ (EJ-204) $\Rightarrow \sigma_{\mathrm{stat}}(t_4)$ intrínseca menor para EJ-230.
- **Sin referencia EXEC_12-EJ230:** el ajuste estilo-EXEC_12 ($25\,\mathrm{ps}$ bins) se computa sobre datos EJ-230 como referencia informativa; no existe publicación previa.
- **Fuera de alcance (bloques posteriores):**
  - Comparación lado a lado EJ-204 vs EJ-230.
  - Introducción de SPTR/jitter (EXEC_02b).
  - MPPC solo en extremos; optimización de espaciado de SiPMs.
  - Corrección de time-walk (HOOK_WALK reservado).
- **G-OVER:** revisar `exec13_230_report.log` para Overfull/Underfull — listado en resumen final.